Telling Exponential Change from Linear Change

Constant differences against constant ratios, evaluating both models, and comparing them over the long run

Directions

- One test settles which model a table follows: over equal steps in the input, a **linear** pattern has equal *differences* between outputs, and an **exponential** pattern has equal *ratios*.
- Linear rules look like $a + bx$; exponential rules look like $a \cdot r^x$, where r is the number you multiply by at each step.
- Round money and population answers to two decimal places where a decimal appears.
- Where a problem asks you to compare two models, use the numbers you computed as the evidence.

1. A table of a function:

x	0	1	2	3
y	5	9	13	17

(a) Subtract consecutive outputs. $9 - 5 =$ _________

(b) Divide consecutive outputs. $9 \div 5 =$ _________

(c) Which model does this table follow, linear or exponential? Justify your choice from the numbers in (a) and (b).

2. A different table:

x	0	1	2	3
y	6	18	54	162

(a) Divide consecutive outputs. $18 \div 6 =$ _________

(b) Subtract the first pair. $18 - 6 =$ _________

(c) Subtract the next pair. $54 - 18 =$ _________

3. Let $f(x) = 4 + 6x$ and $g(x) = 4 \cdot 2^x$. Both start at 4 when $x = 0$.

(a) $f(5) =$ _________

(b) $g(5) =$ _________

4. A town of 200 people is modelled two ways. Model L adds 30 people each year. Model E multiplies by 1.15 each year, a 15% rise.

(a) Population after 4 years under model L: _________

(b) Population after 4 years under model E, to two decimal places: _________

5. A table of a decaying quantity:

x	0	1	2	3
y	80	40	20	10

(a) Divide consecutive outputs. ratio = ________

(b) Continue the pattern. value at $x = 4$: ________

6. Let $f(x) = 50 + 20x$ and $g(x) = 5 \cdot 2^x$.

(a) $f(4) =$ ________ (b) $g(4) =$ ________

(c) $f(8) =$ ________ (d) $g(8) =$ ________

7. Two savings plans each start with \$500. Plan A adds \$40 every month. Plan B grows by 6% every month, so it multiplies by 1.06 each month.

(a) Value of plan A after 12 months, in dollars: _________

(b) Value of plan B after 12 months, in dollars to two decimal places: _________

(c) Which plan is worth more after a year? Explain whether that advantage grows, shrinks, or stays the same as the years go on.

8. Let $g(x) = 3 \cdot 2^x$. Solve $g(x) = 96$. (Write $96 \div 3$ as a power of 2 first.)

Answer: __________________

9. A car bought for \$24,000 is valued two ways. The straight-line method subtracts \$2400 every year. The percentage method multiplies by 0.88 every year, a 12% loss.

(a) Value after 5 years, straight-line method: ________

(b) Value after 5 years, percentage method, to two decimal places: ________

10. Let $f(x) = 100 + 25x$ and $g(x) = 100 \cdot 1.2^x$. They start together at 100.

(a) $f(3) =$ ________ (b) $g(3) =$ ________

(c) $f(4) =$ ________ (d) $g(4) =$ ________

(e) At which whole-number input does g first become the larger of the two? Justify your answer using the four values above.